

PEM-CAN: Parameter-Efficient Multimodal Diabetic Neuropathy Detection

Orthogonal Low-Rank Subspace Cross-Attention over Retinal Fundus & Continuous Wearable Biosignals

1. CLINICAL CRISIS & MODALITIES

The Clinical Blindspot (DAN)

- 2x 5-yr mortality (silent ischemia)
- HbA1c is blind to glycemic volatility
- CARTs reflex labs lack outpatient scale

Ocular & Wearable Windows

- Retina: Microvascular remodeling
- CGM: 5-min interstitial glucose (T=2016)
- Actigraphy: HRV, stress, rest dynamics

Computational Hurdles

- Visual dominance (modality collapse)
- 10%-50% sensor packet dropouts
- Quadratic memory blowup in transformers

2. PEM-CAN SUBSPACE ADAPTERS

Frozen ViT-B/16

M=576 tokens

Dilated Conv1D

CGM + Actigraphy (L=252)

Stiefel LoRA

$\|AA^T - I_r\|_F^2$
Rank $r = 8$
Freeze W_0

Bidirectional Cross-Attn

$S(Z_v, Z_s)$ & $S(Z_s, Z_v)$

- O(M·L) Linear in L
- Anti-Dominance
- Folds into W_0 at Test

$L_{total} = L_{focal} + L_{orth} + L_{align}$

Multi-Task Head

DAN (Autonomic) | DR (Retina)
CKD (Renal) | DPN (Peripheral)

3. BENCHMARK PROTOCOL

AI-READI T2D Protocol

- N = 2,840 Simulated Patients
- Statistically calibrated to NIH atlas
 - All diagnosed Type 2 Diabetes
 - Stratified 5-Fold Cross-Validation
 - African Amer., Hispanic, White cohorts

Non-Leaking Endpoints

- Adjudicated Clinical Labels
- DAN (Autonomic): 22.5% Prev.
 - DR Grade ≥ 2 : 18.2% Prev.
 - CKD Stage ≥ 3 : 14.1% Prev.
 - DPN (Peripheral): 26.8% Prev.

4. QUANTIFIED PERFORMANCE

Diagnostic Discrimination

- AUROC: 0.934 ± 0.008 ($p < 0.001$)
- F1-Score: 0.886 | ECE: 0.038
- Sens @ 90% Spec: 86.4%

Parameter & VRAM Economy

- 2.6M params (1.75% of backbone)
- 8.4 GB training VRAM (vs 32.4 GB)
- 0.861 AUROC under 50% dropout

Edge & Federated Scalability

- Jetson Orin Nano: 20.2 ms / 1.18 GB
- 50-Node FL: 98.25% less bandwidth
- 2.6 MB/round with INT8 quantization